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Classes and objects

FCI224

Learining Objectives

After learning this chapter, the students will be able to

- Understand the purpose of classes, objects Constructors and Destructors
- able to construct C++ programs using classes with Constructors and Destructors
- Execute and debug class programs with Constructors and Destructors

14.1 Introduction to Classes

The most important feature of C++ is the "Class". It is significance is highlighted by the fact that Bjarne Stroustrup initially gave the name 'C with classes'. C++ offers classes, which provide the four features commonly present in OOP languages: **Abstraction, Encapsulation**, **Inheritance**, and **Polymorphism**.

14.1.1 Need for Class

Class is a way to bind the data and its associated functions together. Classes are needed to represent real world entities that not only have data type properties but also have associated operations. It is used to create user defined data type

14.1.2 Declaration of a class

A class is defined in C++ using the keyword **class** followed by the name of the class. The body of the class is defined inside the curly brackets and terminated either by a semicolon or a list of declarations at the end.

Note

The only difference between structure and class is the members of structure are by default **public** where as it is **private in class**.

The General Form of a class definition

```
class class-name
{
private:
        variable declaration;
        function declaration;
protected:
        variable declaration;
        function declaration;
public:
        variable declaration;
        function declaration;
};
```

- The class body contains the declaration of its members (Data member and Member functions).